

Question Paper

Exam Date & Time: 26-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: CSE 3123/CSE_3123 OPERATING SYSTEMS

(SPL: COMPUTER SCIENCE & ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) With a neat diagram, explain the different states of a process. List and explain the different pieces of information stored in Process Control Block. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Explain the different threading models with neat diagram. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Write a note on System boot and bootstrap loader. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) For the given table, apply Banker's Algorithm and perform the following: (5)

Process	Max A B C D	Allocation (A B C D)	Available (A B C D) 3 3 2 2
P1	7 5 3 4	0 1 0 3	
P2	3 2 2 3	2 0 0 1	
P3	9 0 2 2	3 0 2 1	
P4	2 2 2 1	2 1 1 0	
P5	4 3 3 3	0 0 2 2	
P6	3 3 2 2	1 2 1 1	

- Write the content of the matrix Need?
- Is the system safe using Bankers algorithm?
Mention the safe sequence if the system is safe
- Can the request (2,1,1,1) made by process P6 be granted immediately?
- Calculate the total number of instances of each resources type?

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2B) Assume that the following processes are scheduled using Preemptive priority scheduling algorithm. (3)
Assume lower the priority number, highest the priority of the process.

Process	Arrival Time (ms)	Burst Time (ms)	Priority
P2	1	6	7
P4	0	4	2
P3	3	1	0
P6	2	4	3
P5	5	9	0
P1	4	15	1

- i) Show the scheduling order of the processes using ready queue and Gantt chart.
ii) Determine the turnaround time and waiting time for each process.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 2C) Describe with the pseudocode how Test and Set() instruction satisfies all the critical-section problem requirements using bounded-waiting algorithm. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) i) Consider a paging system with the page table stored in memory. (5)
a) If a memory reference takes 50 nanoseconds, how long does a paged memory reference take?
b) If we add TLBs, and 75 percent of all page-table references are found in the TLBs, what is the effective memory reference time? (Assume that finding a page-table entry in the TLBs takes 2 nanoseconds, if the entry is present.)
ii) Describe the concept of segmentation in memory management. Additionally, with a neat labelled diagram, explain the binding of logical address into physical address in segmentation.
(3+2 = 5 marks)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) Discuss the cause of thrashing? How can the system prevent thrashing using working-Set model? (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) Describe what is address binding? Explain three different stages when address binding can be done. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) A process of 10 pages is running in a system that can allocate 4 frames to it. The page reference string is: (5)
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2.
1. Simulate page replacement using Optimal and LRU algorithms.
2. Compute the page fault rate for each algorithm.
3. Based on your analysis, evaluate how page replacement strategy affects system throughput and memory utilization.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Consider a system with 500 cylinders (0 to 499). The head is serving a request at 125 and the previous request served was 140. The queue of pending requests, in FIFO order, is: 85, 120, 20, 35, 5, 7, 135, 10, 400. Starting from the current head position, Calculate the total distance in cylinders that the disk arm moves to satisfy the entire pending request for SCAN and C-SCAN disk (3)

scheduling algorithm. Represent the head movements diagrammatically.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4C) A 4-KB swap slot initially has a reference counter value of 0. (2)
- Process X swaps a page into this slot.
 - Process Y maps to the same page, increasing the reference count.
 - Process X unmaps its reference.
- What will be the final value of the counter for this swap slot, and what does it indicate about the slot's status?

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5A) Consider a system with three domains, D1, D2, and D3, and four files, F1, F2, F3, and F4. The access rights are defined as follows: (5)
- D1: Can read any file and write to F1.
D2: Can read F1, write to F2, and execute F4.
D3: Can write to F1 and F3, and execute F4.
- Domain Switching: A process can switch from domain D1 to D2, D2 to D3, and D2 to D1.
- Construct the access matrix for this system.
 - If a process is in domain D1 and wants to write to file F2, will the operation be allowed? Explain your answer.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) Explain Truncate operation on files. Give names of at least 4 common file types and their extensions. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) Analyse Domain Switch operation in UNIX operating Systems with a suitable example. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name		Subject	
058-END SEMSTER-ODD-2025 CSE_3123:Operating Systems			
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	2
3	3	CO 3	8
4	4	CO 4	15
5	5	CO 5	15
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025 CSE_3123:Operating Systems			
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	7(14%)
3	Application	5	16(32%)
4	Analysis	6	24(48%)
5	Create	1	3(6%)

6	Evaluate	0	0(0%)
	Total	15	50 (100%)